Wireshark Level 3 – Network Traffic Capture & Analysis

This project demonstrates practical use of Wireshark to capture and analyze network traffic for cybersecurity purposes. It focuses on understanding packet structure, capturing live traffic, and performing analysis for threat detection.

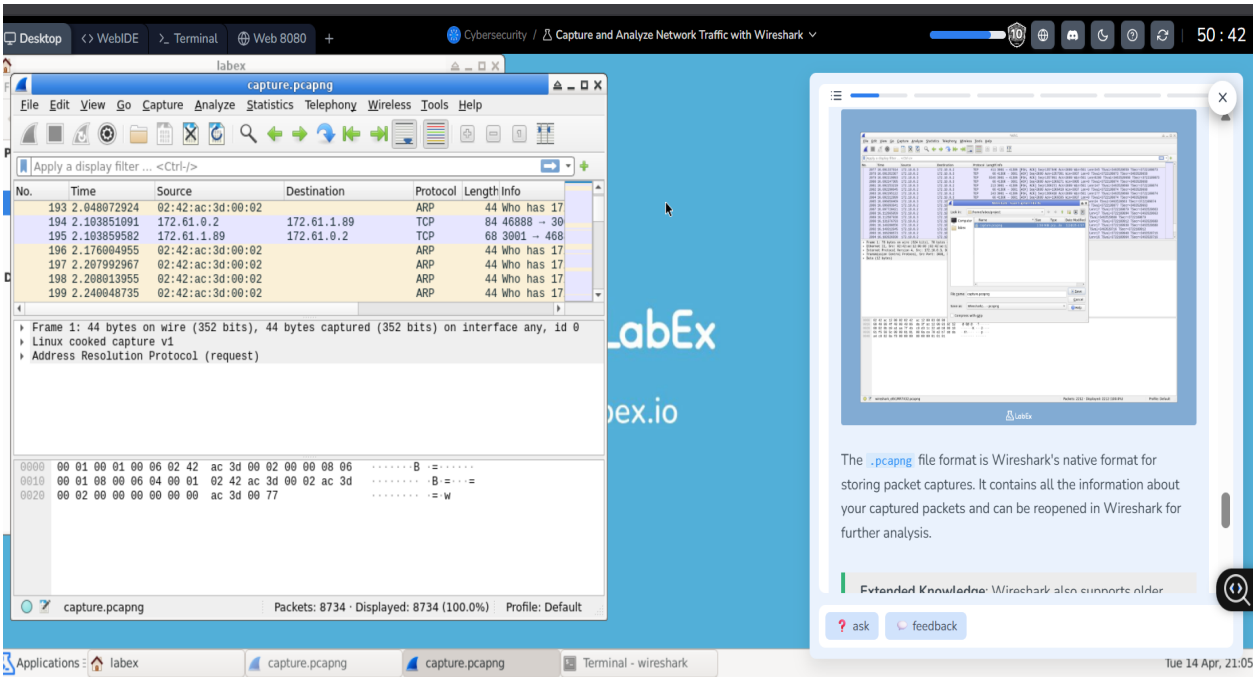
Objective: Capture live network traffic, save it in .pcapng format, and analyze packets to identify communication patterns and potential security threats.

Key Concept – Network Packet:
A network packet is a structured unit of data transmitted over a network. It contains source and destination addresses, payload data, and metadata for routing and processing.

Steps Performed:
1. Launched Wireshark.
2. Selected active network interface.
3. Captured live network traffic.
4. Observed packets including ARP and TCP traffic.
5. Saved capture as .pcapng file.
6. Applied filters and analyzed packet details.
7. Reviewed packet structure and communication patterns.

Analysis Insights:
• ARP packets indicate local network device discovery.
• TCP packets show active communication sessions.
• Source and destination fields help identify communicating hosts.
• Packet lengths and timing reveal network behavior patterns.

Importance of Statistical Analysis:
Statistical tools in Wireshark help identify anomalies, monitor bandwidth usage, detect intrusions, and establish normal network baselines.



Conclusion:
This lab demonstrates essential packet analysis skills required in cybersecurity. The ability to capture and analyze traffic enables detection of suspicious activity, troubleshooting, and network monitoring.